pfi

Python for Fixed Income

COURSE 2 · BOND MATH

Spread Duration and DTS

WEEK 3 · DAY 3

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

From a spread to its sensitivity

Day 12 measured each bond's spread to the curve, a level. Today: if that spread moves, how far does the price move?

1. **Spread duration: bump the spread 1 bp each way**
2. **Why it equals modified duration here**
3. **DTS: spread duration times spread**
4. **The file's two columns reproduced**
5. **The book's DTS by bucket and by bond**

DTS is a published measure. Every table describes the synthetic book; nothing says how much a book should carry.

Week 3: curve, spread, and the book

Day 11	Treasury curve interpolation
Day 12	Spread over the curve
Day 13	**Spread duration and DTS**
Day 14	Aggregation and scenarios
Day 15	Capstone

Spread duration by bumping the spread

```
# 1 bp is 0.01 in yield_pct (percent)
down = bondmath.price(s, m, c, y - 0.01)
up = bondmath.price(s, m, c, y + 0.01)
dirty = bondmath.dirty_price(s, m, c, y)
# and 0.0001 in the slope (decimal)
(down - up) / (2 * 0.0001) / dirty
```

Bezane Finance, 99080CAE8	Value
Clean price, spread 1 bp lower	106.776909
Clean price, spread 1 bp higher	106.676310
Difference over 2 bp	0.100600
Spread duration, years	4.626971

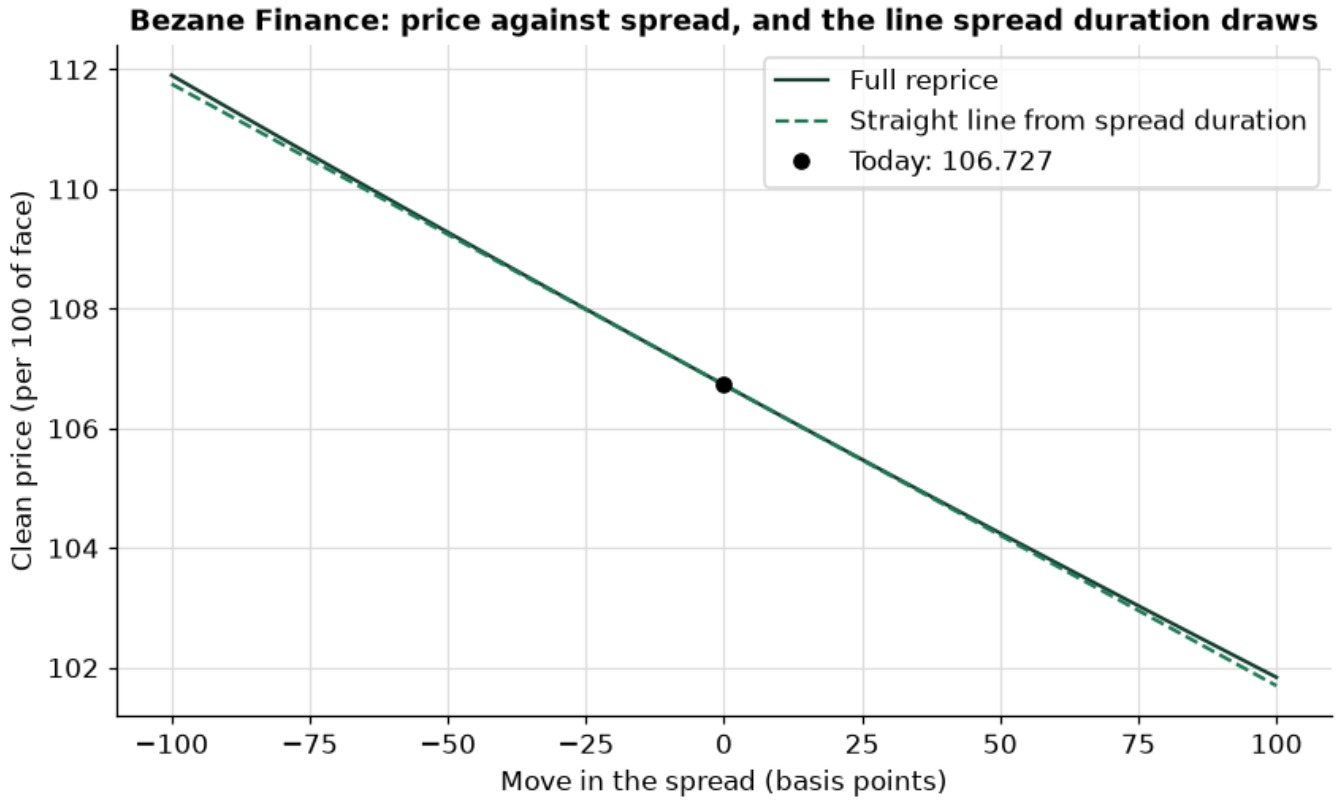
Excel side by side (tested in Excel)

`B1: =PRICE(...,5.606/100,100,2,0)`	106.726595
`B2: =PRICE(...,5.606/100-0.0001,...)`	106.776909
`B3: =PRICE(...,5.606/100+0.0001,...)`	106.676310
`B4: the accrued formula`	1.983333
`B5: =(B2-B3)/(2*0.0001)/(B1+B4)`	4.626971

One bump, two spellings: 0.01 in percent, 0.0001 as a decimal.

All 200 holdings bonds match Excel's bumped PRICE columns, largest gap 3.0e-11 years.

The line spread duration draws



4.626971

Bezane's spread duration, years

The solid line is a full reprice. The dashed line is today's price moved by spread duration alone.

They part slowly because the price curve bends. For a few bp they cannot be told apart.

Why it equals modified duration here

```
yield = curve yield + spread / 100
```

Bezane: curve 4.136 + 147 bp / 100 = 5.606 percent. Hold the curve still: a 1 bp spread move is a 1 bp yield move.

So the slope against spread is modified duration, from day 9.

1.61e-05

largest gap to modified duration over 200 bonds, years

5e-5

the course's tolerance for the test

Sylvellix Logistics, modified duration 16.042

Bump, bp	Spread duration	Minus modified
1	16.042490	1.61e-05
5	16.042875	4.02e-04
25	16.052514	1.00e-02
100	16.203732	1.61e-01

A bump is a straight line between two points of a curved price line. Five times the bump, about twenty-five times the gap.

A bond in its final coupon period differs: Excel prices it with simple interest. The test skips those rows.

DTS: spread duration times spread

```
DTS = spread duration x spread   (years x bp)
```

A spread that moves by a share of itself, k: price change in percent = -spread duration x (k x spread) / 100 = -DTS x k / 100.

680.16

Bezane's DTS, years x bp

-0.6802%

DTS estimate, spread 14.7 bp (10%) wider

-0.6773%

full reprice, the same move

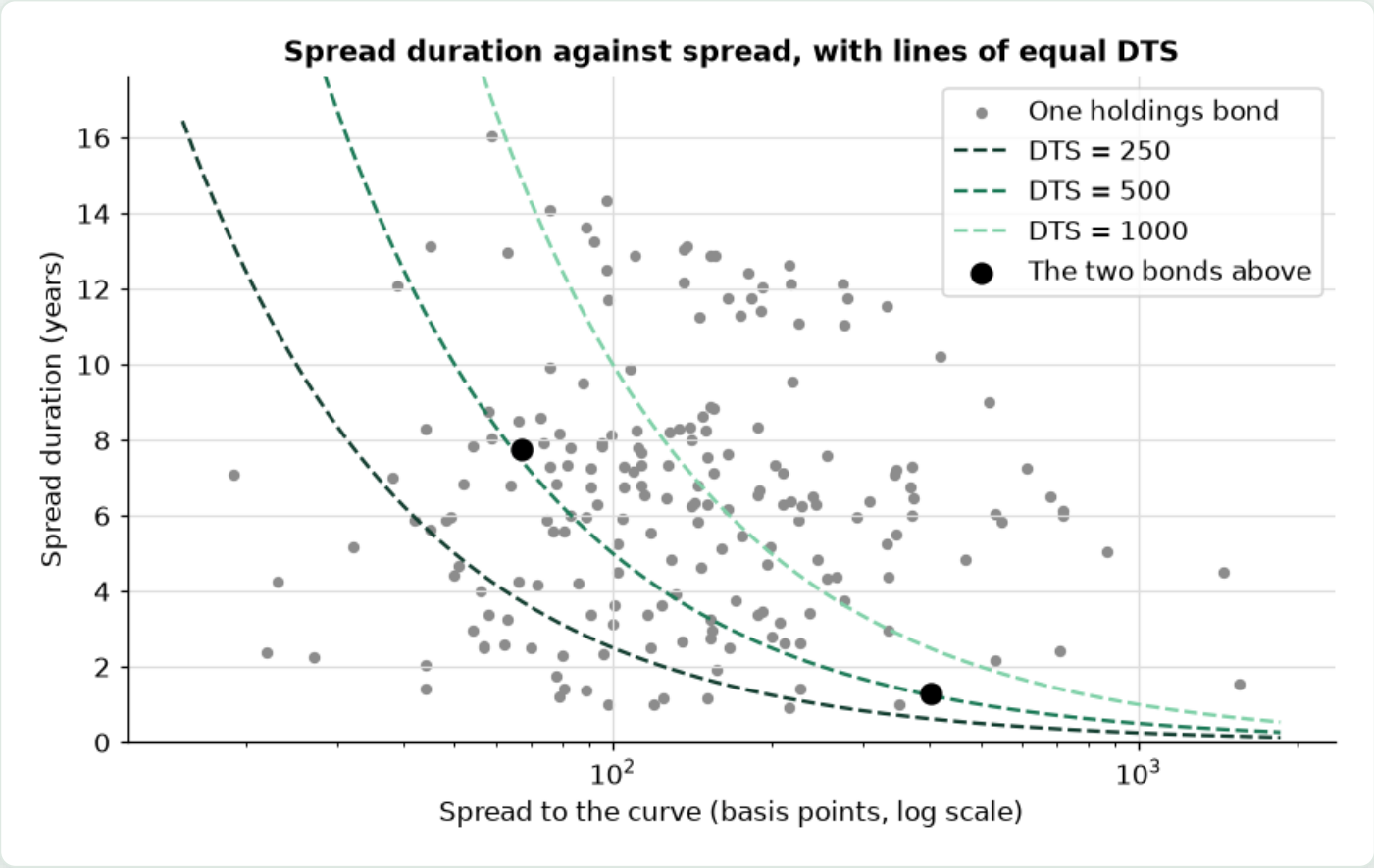
Arithmetic on one invented bond, not a forecast of any spread. The small gap is convexity.

Two bonds of the book, close DTS

	Galvellix Paper	Zephizal Medical
Rating	B-	A-
Spread, bp	401	67
Spread duration	1.310	7.743
DTS	525.5	518.8

Two routes to one DTS. It says nothing about holding either.

Lines of equal DTS



Each dashed curve is one DTS: 250, 500, 1,000. A bond on a curve has that DTS, however it got there.

The two black dots have close DTS: one has a wide spread, the other a long duration.

The spread_duration and dts columns reproduced

The file set spread_duration = duration, the modified duration rounded to 3 decimals, and dts = that times oas, rounded to 1 decimal.

0.000498707

largest spread duration gap, 99010KAH8: 5.950498707 against 5.95

0.3999

largest DTS gap from the bond math alone (a spread of 871 bp)

0.05

largest gap, the file's dts from its own inputs

Column	Tolerance	Why
spread_duration	0.0005 + 1e-9	rounded to 3 decimals
dts, from the file's inputs	0.05 + 1e-9	rounded to 1 decimal

37 passed: 35 from days 1 to 12, and today's two.

Weights and contributions

```
weight_pct = market_value / total * 100
dts_contribution = weight_pct / 100 * dts_bm
book_dts = dts_contribution.sum()
# = (market_value * dts_bm).sum() / total
```

Excel side by side (tested in Excel)

`L2: =(I2-J2)/(2*0.0001)/(H2+K2)`	spread duration
`M2: =L2*F2`	DTS
`N2: =G2/SUM($G$2:$G$201)*100`	weight
`O2: =N2/100*M2`	contribution
`=SUMPRODUCT(G2:G201,M2:M201)/SUM(G2:G201)`	1064.11

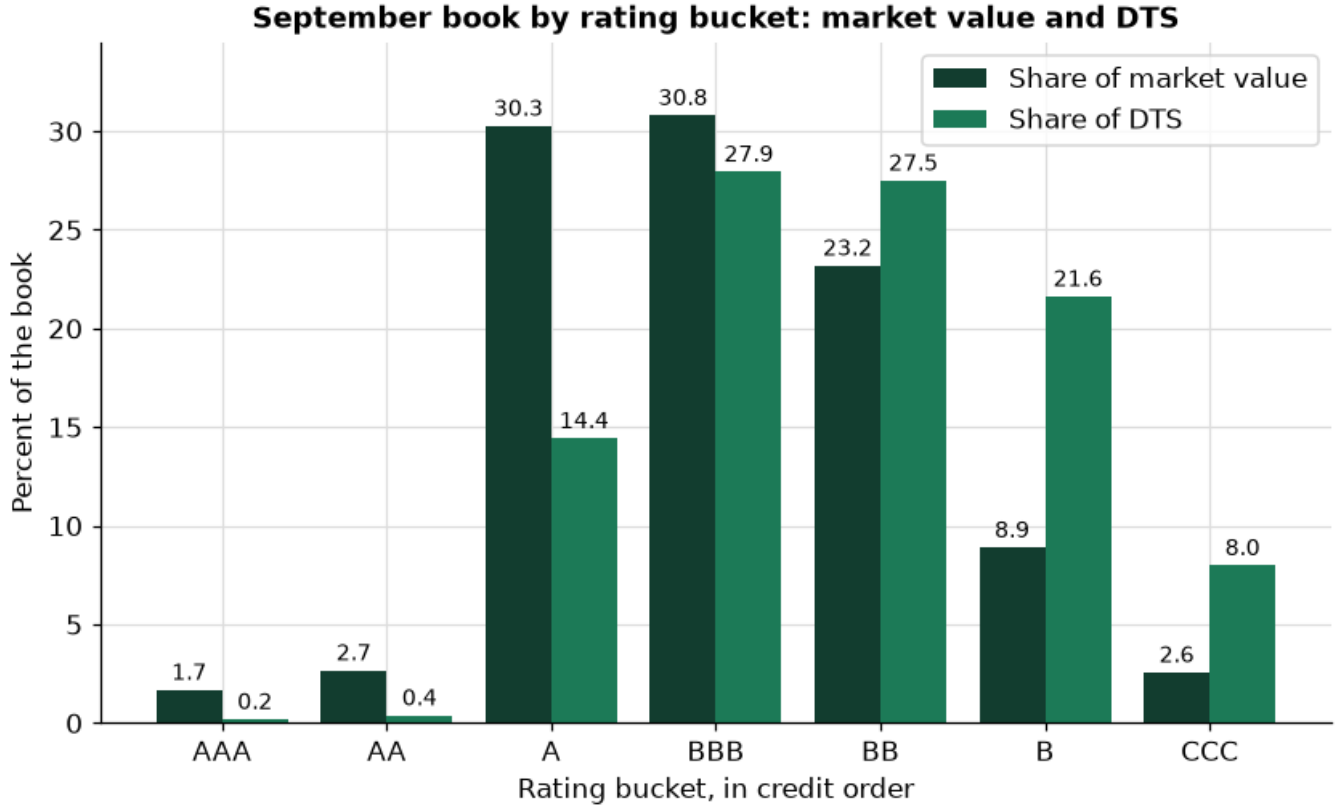
1064.11

book DTS, years x bp, every input our own

Day 14 rebuilds market_value from par and the dirty price. Today it is read from the file.

=SUMIFS(O2:O201,B2:B201,"BBB") returns 297.24 for BBB.

Market value and DTS by bucket



34.62%

High Yield: share of the value

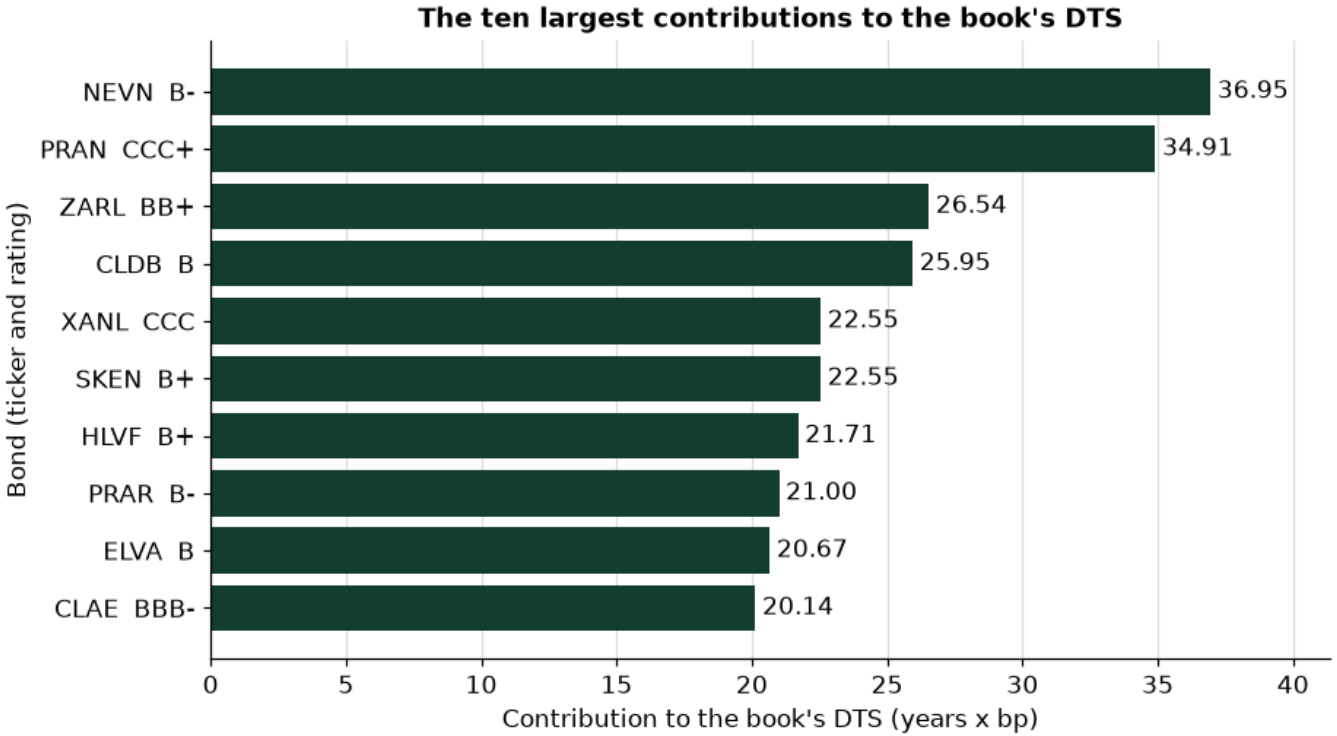
57.09%

High Yield: share of the DTS

The dark bar is a bucket's share of market value, the green bar its share of DTS. One crossing, going down the scale.

A description of an invented book, not a view on any bucket.

The ten largest contributions



23.8%

of the book's DTS

7.96%

of its market value

Weight times DTS, bond by bond. A bond gets onto the list through a wide spread, a long spread duration, a large weight, or some of each.

A description of the book, not a view on any bond.

Today's Excel, every formula tested

Job	Excel (tested), Bezane	Returns
Price, spread 1 bp lower	`=PRICE(DATE(2026,9,30),DATE(2032,6,18),7/100,5.606/100-0.0001,100,2,0)`	106.7769
Spread duration	`=(B2-B3)/(2*0.0001)/(B1+B4)`	4.626971
Modified duration	`=MDURATION(DATE(2026,9,30),DATE(2032,6,18),7/100,5.606/100,2,0)`	4.626971
Curve plus spread	`=PRICE(...,(4.136+(147+1)/100)/100,100,2,0)`	106.6763
DTS	`=B5*147`	680.16
Book DTS	`=SUMPRODUCT(G2:G201,M2:M201)/SUM(G2:G201)`	1064.11
One bucket	`=SUMIFS(O2:O201,B2:B201,"BBB")`	297.24

Inside PRICE the yield is a decimal: 1 bp is 0.0001. In bondmath, 0.01 in yield_pct and 0.0001 in the slope.

git log -S: when did a number first appear?

```
git log -S "10000" --oneline
```

The pickaxe lists the commits that changed how many times the text appears: the commits that added or removed it. It searches the changes, not the messages.

The work folder is new: the notebook first rebuilds one commit per day, days 1 to 12.

What comes back

```
HASH Add spread_duration and dts, ...
HASH Day 9: Duration and DV01
```

Day 9's dv01 docstring: =MDURATION(...)*(PRICE(...)+accrued)/10000. The oldest commit listed is where the number first appeared.

```
git log -S "10000" --oneline -- test_bondmath.py
```

Limited to one file: only day 9. A name works too: -S "bump_decimal" finds today.

Bring it home: paste, test, commit, search

1

Your folder

Set-Location to my_bondmath, in your projects folder.

2

Paste

Today's two functions (13.1, 13.3) and two tests (13.2, 13.4), without the %%writefile line.

3

Test

python -m pytest -q: 37 passed, or more with your exercise functions.

4

Commit

git diff --stat, git add, git commit with a subject and a body.

5

Search

git log -S "10000" --oneline on your own history.

In Colab: download bondmath.py and test_bondmath.py from the work folder under /tmp. They replace yesterday's downloads.

The AI check

The docstring is the prompt. The code that comes back runs, and has one convention error.

1

Docstring

Units, convention, and the Excel formula: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert against Excel's reference rows. Run it: it fails.

4

Fix

Find the convention it broke, fix it, run the test again.

Today: the slope is divided by the bump in bp, not as a decimal, so spread duration is 10,000 times too small.

Code that runs is not the same as code that is right.

What you can do now

- Measure spread duration by bumping the spread, in Python or Excel
- Say why it equals modified duration here, and where it differs
- Compute DTS and say what it measures
- Reproduce the spread_duration and dts columns
- Add up the book's DTS by bucket and bond; find a commit with git log -S

Week 3, Day 4 Portfolio aggregation and scenarios: the whole book, then rates and spreads moved